Computer Systems Architecture and Operating Systems

CST1500

17/10/2024

Batch Scripting

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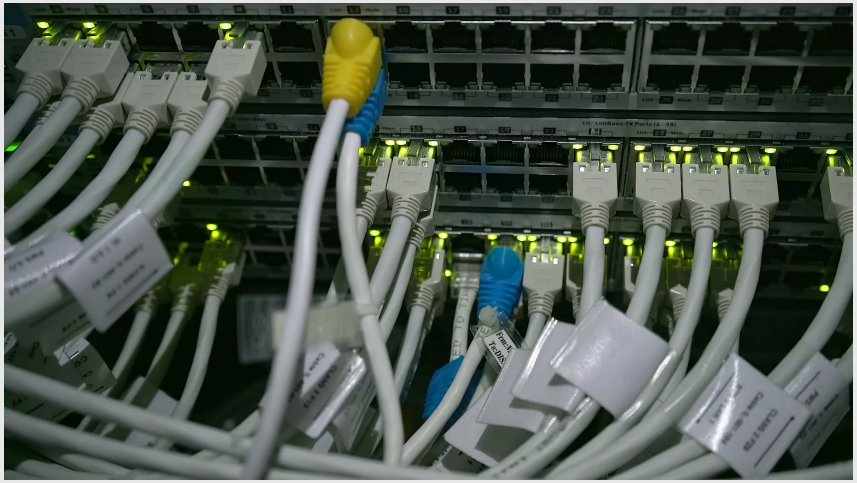
Workshop: Week 4

17/10/2024



Plan for Today

- Overview of Weekly Topics
- Introduction to Batch Scripting
- Why Do We Need Batch Scripting?
- Batch Scripting Commands
- Code Demonstration
- SOB6, SOB7, SOB8 : 15% of Final Mark
- Summary

Learning Outcomes

- Understand the Basics of Bash Scripting and Its Structure.
- Identify the Importance and Applications of the Bash Scripting Language.
- Write and Execute Bash Scripts Effectively.
- Demonstrate the Use of Bash Scripting in Real-World Applications.

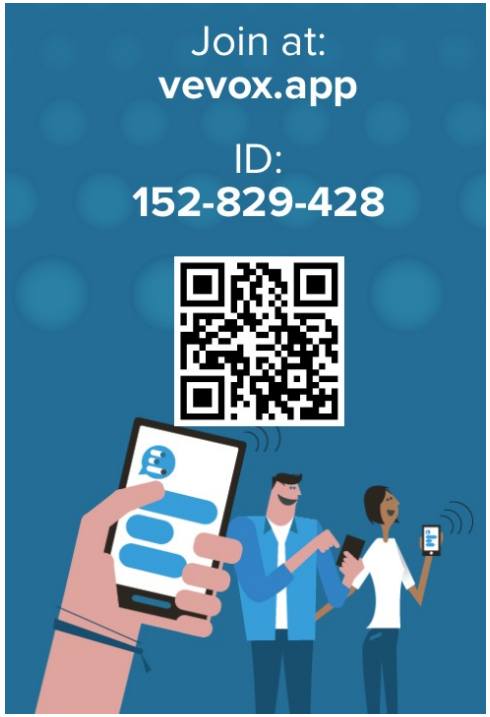
Overview

- Week 1-3:
  - Fundamental of Computer Systems
  - C Programming
- Week 4: Batch Scripting Language
- Week 5: Bash Scripting Language

- Week 6: CW1: Submission of SOBs (50%)
- Weeks 7–11: Python and Arduino
- Week 12: CW2: Final Project (50%)

Introduction to Batch Scripting

Have you ever used Batch Scripting?



Introduction to Batch Scripts

- A **shell** is a command-line interpreter that reads and executes commands entered by the user.
- It acts as a bridge between the user and the operating system, interpreting user commands and executing them.
- **Batch**: The shell scripting language used in **Windows**.
- **Bash**: A popular shell scripting language for **Linux**.
- Each shell has its own syntax and features.
- **Batch scripts** are executed via **CMD (Command Prompt)** on Windows.

Why Do We Need Batch Scripting?

- Batch scripting is used in Windows environments to:
  - automate repetitive tasks
  - enhance productivity
  - manage system operations

Why Do We Need Batch Scripting?

- **Some Real Applications of Batch Scripting:**

1. Task Automation

- Batch scripts can automate repetitive tasks such as file management, software installation, and system maintenance.
- This saves time and effort, especially for tasks that need to be performed frequently or on multiple machines.

Real Applications of Batch Scripting

2. System Administration

- System administrators can use batch scripts streamline administrative tasks, such as backups, log cleanup, and software updates.

3. Scheduling Tasks

- Batch scripts can be used with Windows Task Scheduler to schedule tasks to run at specific times or intervals. This is useful for tasks like automated backups, system monitoring, or routine maintenance.

Real Applications of Batch Scripting

4. Environment Configuration

- Batch scripts can configure the environment by setting environment variables.

• 5. File and Process Management

- Batch scripts can automate file operations such as copying, moving, renaming, or deleting files. They can also start, stop, or monitor processes, making them useful for managing services and applications.

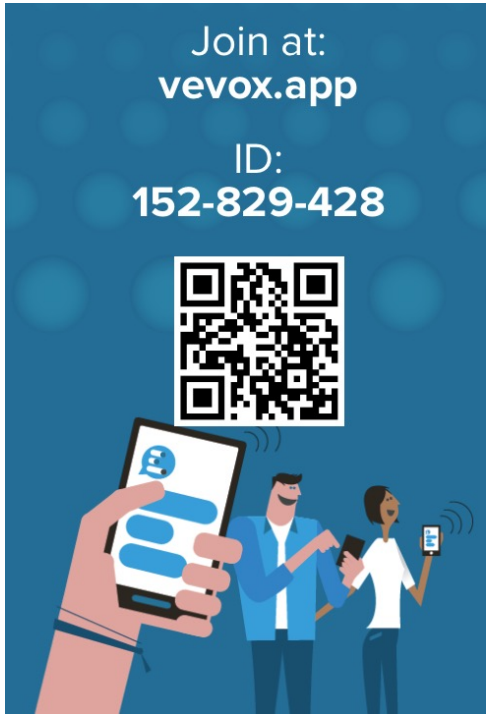
Real Applications of Batch Scripting

6. Easy Troubleshooting

- Batch scripts can be written to gather system information, perform diagnostic checks, and provide reports.

- Note:
  - PowerShell: OOP developed by Microsoft, designed specifically for system administration and automation.

Who Should Learn Batch Scripting?

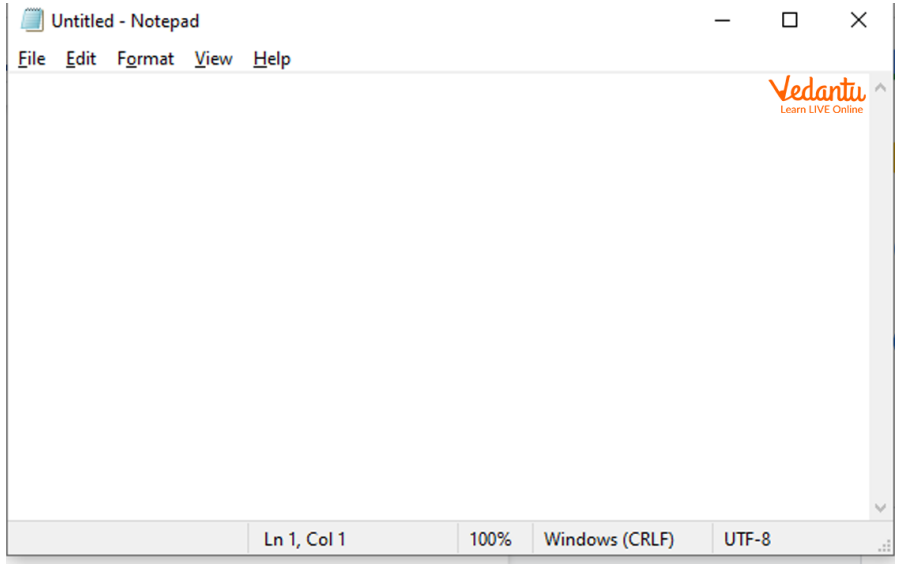


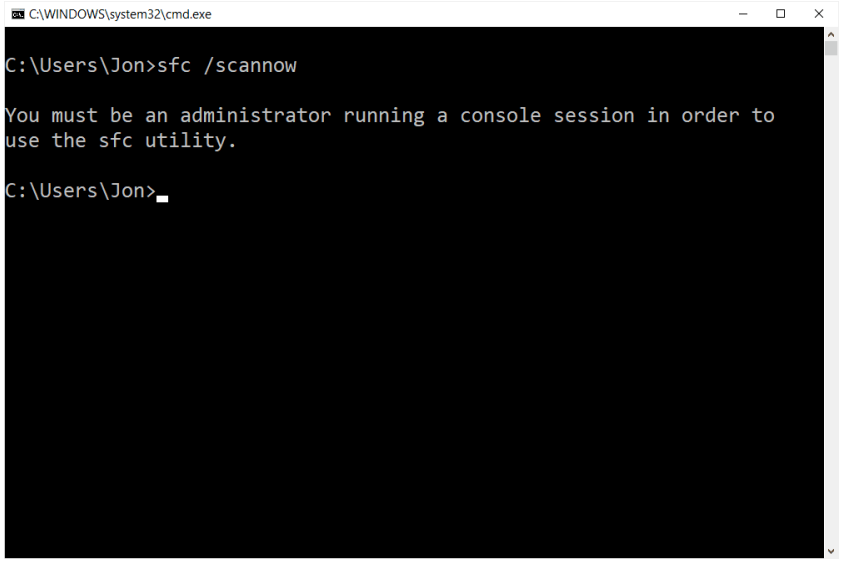
Who Should Learn Batch Script?

- **System Administrators**
  - Automate routine administrative tasks and manage servers efficiently.
- **Software Developers**
  - Automate testing.
- **IT Support Specialists**
  - Simplify support tasks and enhance troubleshooting processes.
- **Automation Engineers**
  - Design and implement automated solutions for complex processes.

Writing and Executing

- To create a batch file, notepad is used.
- Command prompt (Terminal) is used as the execution environment for the batch scripts.





Commands

- cd
  - CD stands for "Change Directory." It is a command used to change the current working directory in the command prompt or within a batch file.
- cls
  - This batch command clears the screen.
- copy
  - This batch command is used for copying files from one location to the other.
- del
  - This batch command deletes files and not directories.

Commands (Cont.)

- dir
  - lists the contents of a directory.
- echo
  - displays messages or turns command echoing on or off.
- exit
  - exits the DOS console.
- mkdir
  - creates a new directory in the current location.
- move
  - moves files or directories between directories.
- pause
  - prompts the user and waits for a line of input to be entered.
- rmdir
  - remove an empty folder.

Commands (Cont.)

- rem
  - remarks in batch files, preventing the content of the remark from being executed.
- start
  - starts a program in new window or opens a document.
- type
  - prints the content of a file or files to the output.
- title
  - sets the title displayed in the console window.

Create a folder

cmd Copy code

```
cd %USERPROFILE%\Desktop
```

cmd Copy code

```
mkdir MyFolder
```

Create a file

- Create an empty file:

cmd Copy code

```cmd
type nul > emptyfile.txt
```

- Create a file with content:

cmd Copy code

```cmd
echo Hello, World! > example.txt
```

Copy and Move

cmd Copy code

```cmd
copy example.txt MyFolder
```

cmd Copy code

```cmd
move example.txt MyFolder
```

List the files and folders

cmd

```
dir
```

```
Directory of C:\Users\YourUsername\Desktop

10/15/2024  10:00 AM    <DIR>          MyFolder
10/15/2024  10:00 AM                1,024 example.txt
10/15/2024  10:00 AM    <DIR>          AnotherFolder
10/15/2024  10:00 AM                2,048 document.docx
               2 File(s)          3,072 bytes
               2 Dir(s)  123,456,789 bytes free
```

cmd

```
dir /b /s
```

```
C:\Users\YourUsername\Desktop\MyFolder
C:\Users\YourUsername\Desktop\MyFolder\SubFolder
C:\Users\YourUsername\Desktop\example.txt
C:\Users\YourUsername\Desktop\AnotherFolder
C:\Users\YourUsername\Desktop\AnotherFolder\document.docx
```

run and remove

- Run a file

```cmd
example.bat
```

- Remove an empty folder

```cmd
rmdir MyFolder
```

- Remove a folder

```cmd
rmdir /s MyFolder
```

- Delete only the files in a folder

```cmd
del *.*
```

Batch Scripts

- Create a Batch file named **hello.bat** using Notepad.

```
@echo off
echo Hello, World!
pause
```

```
Hello, World!
Press any key to continue . . .
```

Batch Scripts

test.bat

```
@echo off
echo %1
echo %2
echo %3
```

```
C:\tp>test.bat 1 2 3
```

Output?

```
1
2
3
```

set variables

```
@echo off
set message=Hello World
echo %message%
```

```
Hello World
```

Four operations

```
@echo off
SET /A a = 5
SET /A b = 10
SET /A c = %a% + %b%
echo %c%
SET /A c = %a% - %b%
echo %c%
SET /A c = %b% / %a%
echo %c%
SET /A c = %b% * %a%
echo %c%
```

Output:

```
15
-5
2
50
```

Comments

- REM: remark

batch Copy code

```batch
@echo off                    REM Turn off command echoing
echo Hello, World!          REM Display a greeting message
pause                       REM Wait for the user to press a key
```

For loop

```
@echo off
set list=1 2 3 4
(for %%a in (%list%) do (
   echo %%a
))
```

```
1
2
3
4
```

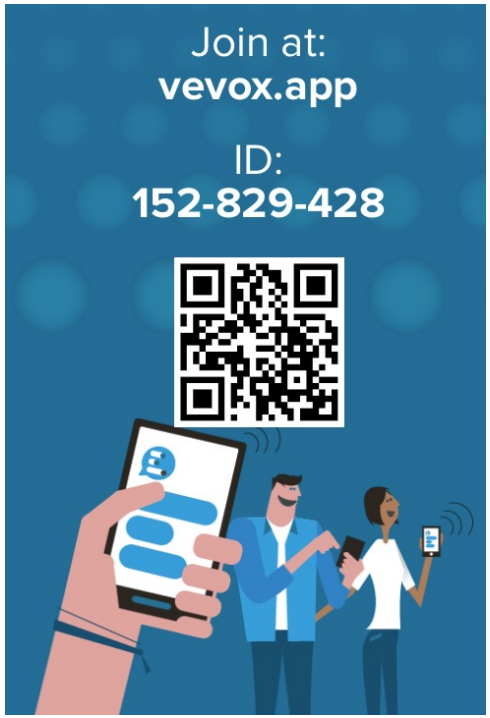
For loop (cont.)

```
@echo off
REM Loop from 1 to 5
for /l %%i in (1, 1, 5) do (
    echo Number: %%i
)
pause
```

```
Number: 1
Number: 2
Number: 3
Number: 4
Number: 5
Press any key to continue . . .
```

What is the purpose of this code?

```
@echo off
for %%f in (*.txt) do (
    echo File found: %%f
)
pause
```



Built-in variables

```
@echo off
echo Hello, World!
echo This is a simple Batch script.
echo Today's date is: %date%
echo The current time is: %time%
pause
```

```
Hello, World!
This is a simple Batch script.
Today's date is: Mon 10/15/2024
The current time is: 12:34:56.78
Press any key to continue . . .
```

What else ?

While – if

```
@echo off
SET /A "index = 1"
SET /A "count = 5"
:while
if %index% leq %count% (
   echo The value of index is %index%
   SET /A "index = index + 1"
   goto :while
)
```

```
The value of index is 1
The value of index is 2
The value of index is 3
The value of index is 4
The value of index is 5
```

Operator	Description	Example
`==`	Equal to	`if %var% == 10`
`!=`	Not equal to	`if %var% != 10`
`gtr`	Greater than	`if %var% gtr 10`
`lss`	Less than	`if %var% lss 10`
`geq`	Greater than or equal to	`if %var% geq 10`
`leq`	Less than or equal to	`if %var% leq 10`

Another Example

```
@echo off
SET /A "index=1"
SET /A "count=5"
:while
if %index% leq %count% (
   if %index%==2 goto :Increment
      echo The value of index is %index%
:Increment
   SET /A "index=index + 1"
   goto :while
)
```

```
The value of index is 1
The value of index is 3
The value of index is 4
The value of index is 5
```

if statement

```
@ECHO OFF
:Loop

IF "%1"== GOTO completed
FOR %%F IN (%1) DO echo %%F
SHIFT
GOTO Loop
:completed
```

test.bat 1 2 3

```
1
2
3
```

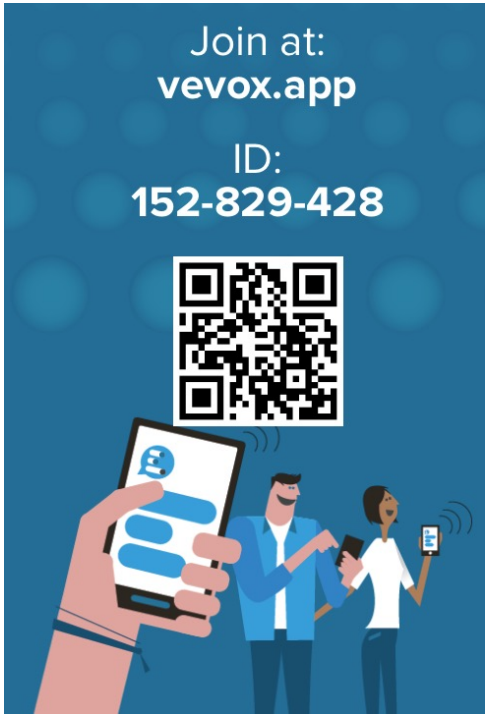
shift

```batch
@echo off

:: Check if at least one argument is provided
if "%1"==="" (
    echo No arguments provided.
    exit /b
)

echo Processing arguments:
:loop
echo %1  :: Display the first argument
shift   :: Shift parameters to the left

if not "%1"==="" goto loop  :: Repeat until no arguments remain

echo All arguments processed.
pause
```

```
shift_example.bat arg1 arg2 arg3
```

```
Processing arguments:
arg1
arg2
arg3
All arguments processed.
Press any key to continue . . .
```

What is the output?




```
@echo off


set /p n="Enter a positive integer (n): "


set /a x=0


if not defined n (
    echo You must enter a number.
    goto end
)

rem Loop from 1 to n
for /L %%i in (1,1,%n%) do (
    set /a x+=%%i
)

:end
echo The output is: %x%


pause
```

n=10

Code Demo: Factorial

5! (Factorial of 5):

5!=5×4×3×2×1=120

```
@echo off

if "%1"==""  (
    echo Please provide a number:
    exit /b
)

set /a number=%1
set /a result=1

:factorial
if %number% lss 2 (
    goto eof
)

set /a result=result * number
set /a number-=1
goto factorial

:eof
echo The factorial of %1 is %result%
pause
```

```
factorial.bat 5
```

Output: The factorial of 5 is 120

Conclusion

- Batch scripting is a powerful and efficient way to automate tasks in Windows operating systems.
- **Basic Structure**: Understanding the fundamental syntax and commands, such as echo, set, dir, rmdir, etc. enables us to create simple scripts for various tasks.
- **Control Flow**: Utilizing loops (for), conditional statements (if), and functions enhances the capability of scripts, allowing for more complex and dynamic automation processes.
- **Practical Applications**: Batch scripts can be employed in real-world scenarios, such as automated backups, etc.

SOBs: Batch Scripts

- SOB6: 5%
- SOB7: 5%
- SOB8: 5%